CCCA Meta-Atlas — Griffiths et al. 2021 (Breast)

Breast Cancer (ER+ neoadjuvant) · 10x + iCell8 · 91,802 malignant cells analysed (Body compartment; 74 samples ≥50)

Do malignant cells recapitulate embryonic programs? Non-branching (linear) manifold — GO / GO / NA

GO / NO-GO GATES

Gate 1 — mixing & continuity: GO

Harmony (θ=10) on sample integrates 74 samples across two technologies. Mixing is judged on the per-neighbourhood iLISI (eval-spec metric), median 1.02→11.71 — not on the coarse same-sample-neighbour scalar (which is ~flat across θ=2/5/10). The manifold resolves into a large Body (91.8k) and a low-complexity left lobe; the Body reproduces the reference Gast↔pSom cross-repression and is one continuous mass (single kNN-graph component; 96% coherent diffusion-map mass) → GO, scoped to the Body.

Gate 2 — module geography: GO (non-branching)

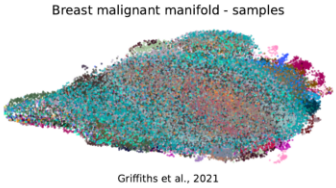
Organized Gast/pSom cross-repression (smoothed r=−0.45). Along pseudotime pSom falls (ρ=−0.65) while Crest rises (ρ=+0.57); NC-high decile at t=0.85 (terminal). Joint pSom&Crest enrichment 0.31<1 → inverse terminal gradient, a single (Gast/pSom)→Crest axis. Present but not resolved into two arms → GO. FLAG: Crest co-enriches the Gast pole marginally more than pSom (0.33 vs 0.31) — arm inversion flagged, pass anyway.

Gate 3 — cross-repression: NA

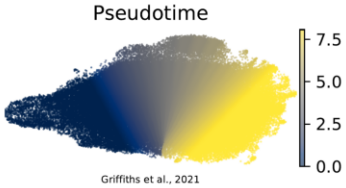
Non-branching topology (principal curve: 2 tips, 0 forks across N=15/20/30; fold-free self-approach 2.86). No bifurcation to test → NA (branching-only correlation/modules notebooks not produced).

STEP FIGURES

Samples



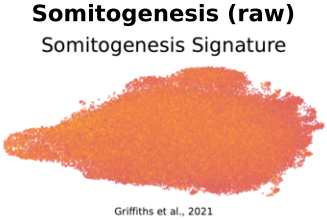
Pseudotime



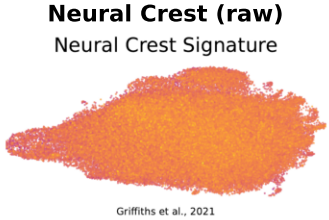
Gastrulation (raw)



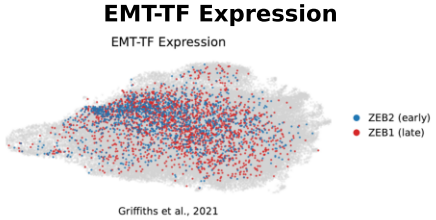
Somitogenesis (raw)



Neural Crest (raw)



EMT-TF Expression



PARAMETER CHOICES & RATIONALE

Threshold: ≥50 malignant cells/sample (re-derived). No dominant count cliff (argmax gap sits in the single-cell tail); floor guarantees a stable per-sample Harmony centroid at 10x scale. 77/95 samples, 110,999 cells; the analysed Body is 91,802 of these.

Compartment: Scoped to the Body lobe. The separated left lobe (16,545) is a low-complexity technical partition (~22% fewer genes/cell, low cycling, top markers led by nuclear lncRNAs NEAT1/MALAT1) — excluded.

Embedding: Harmony θ=10 on sample — strongest integration that keeps the Body a single continuous mass (also removes the 10x/iCell8 technology batch); n_pcs=30, n_neighbors=30, UMAP spread=1.3; 4,885 overdispersed genes; no sc.pp.scale before PCA.

Trajectory: scFates principal curve (non-branching), Nodes=20, epg_lambda=0.3, epg_mu=100; fold-free (SA 2.86).

Root: Gast/pSom-high, Crest-low curve tip (developmental origin); corr(x_lin, pseudotime)=+0.94. Axis is not proliferation-driven (pseudotime vs cycling ρ=+0.07).

Geometry: Linear (pSom origin → Crest terminus). Raw module means: Gast −0.020, pSom −0.053, Crest +0.016.

EMT-TFs: ZEB2 (early/root) ↔ ZEB1 (late/terminus), detection-gated from a 10-TF panel incl. FOXC/PRRX families (VIM excluded — a mesenchymal marker, not a TF). ZEB1 9.3% / ZEB2 5.3% expressing.

Verdict: The embryonic pattern reproduces as a single gastrulation/somitogenesis→neural-crest axis (non-branching), with organized Gast/pSom cross-repression (r=−0.45) and a terminal Crest gradient; no Gast↔pSom bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the other CCCA breast datasets. This resolves the prior science failure (gates never determined) to a clean GO / GO / NA call.